

1.

```
SELECT c.customer_name,c.city,SUM(o.amount) AS total_purchase_amount from customer c
JOIN orders o ON c.customer_id = o.customer_id
GROUP by c.customer_id, c.customer_name, c.city
HAVING sum(o.amount) =
(select MAX(city_totals.total) from (
select SUM(o2.amount) AS total FROM customer c2
join orders o2 ON c2.customer_id = o2.customer_id where c2.city = c.city group by
c2.customer_id) city_totals);
```

	customer_name character varying (50) 🔒	city character varying (50) 🔒	total_purchase_amount bigint 🔒
1	Karan	Pune	4500
2	Priya	Mumbai	8000
3	Aman	Delhi	7000
4	Raj	Mumbai	8000

2.

```
create or replace FUNCTION change()
RETURNS TRIGGER
LANGUAGE plpgsql
AS $$
BEGIN
    insert INTO product_audit VALUES (1, NEW.product_id, NEW.product_name, 'PRODUCT
ADDED');
RETURN new;
END;
$$;
```

```
CREATE OR REPLACE FUNCTION change2()
RETURNS TRIGGER
```

```

LANGUAGE plpgsql

AS $$

BEGIN

    INSERT INTO product_audit VALUES (1, OLD.product_id, OLD.product_name, 'PRODUCT
REMOVED');

    RETURN OLD;

END;

$$;

```

```

CREATE TRIGGER insert_trigger
AFTER insert ON product_inventory
FOR EACH ROW
EXECUTE FUNCTION change();

```

```

create TRIGGER delete_trigger
after delete ON product_inventory
FOR EACH ROW
EXECUTE function change2();

```

```

insert into product_inventory values(101,'laptop',65000);

select * from product_audit;

```

	audit_id integer 	product_id integer 	product_name character varying (50) 	action character varying (50) 
1	1	101	laptop	PRODUCT ADDED